

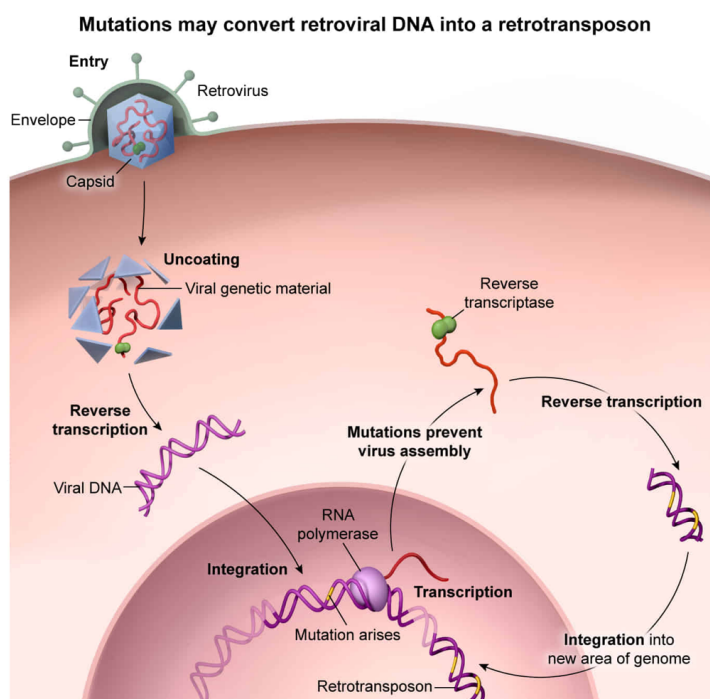
Autonomous retrotransposons are segments of DNA that encode specialized enzymes. These enzymes allow autonomous retrotransposons to move between different regions of a host genome via RNA intermediates using mechanisms similar to those of retroviruses. Some autonomous retrotransposons are segments of retroviral DNA that have been partially inactivated by mutations. Given this, which of the following mutations would most likely convert active retroviral DNA into an autonomous retrotransposon?

- ☐ A. A mutation that prevents the expression of functional reverse transcriptase [29%]
- ☐ B. A mutation that prevents integration of retroviral DNA into the host genome [23%]
- ✓ ☒ C. A mutation that prevents the expression of envelope proteins [21%]
- ☐ D. A mutation that prevents RNA polymerase from transcribing retroviral DNA [26%]

Correct

21% answered correctly

## Explanation:



**Retroviruses** are a class of **enveloped** viruses that have RNA genomes. Upon entering a host cell, the envelope and capsid disassemble, releasing viral RNA and proteins into the cytoplasm of the host cell. Viral **reverse transcriptase** converts viral RNA into double-stranded (ds) DNA that is then transported into the nucleus. A specialized enzyme called integrase facilitates the integration of the viral DNA into the host genome. Once integrated, viral DNA is **replicated** along with the host genome, and **transcription** and **translation** of viral DNA produce RNA and proteins that are assembled into new viruses.

The question states that **autonomous retrotransposons** move via RNA intermediates using mechanisms similar to those of retroviruses. In general, transposons are segments of DNA that move between different areas of the genome and can be classified as either **DNA transposons** or retrotransposons. The retrotransposon is initially transcribed into mRNA, which is then transported into the cytoplasm and translated to produce reverse transcriptase and integrase. Reverse transcriptase *produces dsDNA* from the mRNA, and integrase *facilitates the integration of the dsDNA* into a new area of the genome.

According to the question, some autonomous retrotransposons are segments of retroviral DNA that have been partially deactivated by mutations. Because fully functional retroviral DNA must encode the protein components necessary to assemble new viruses, a **mutation preventing the expression of envelope proteins** would *stop* the production of infectious retroviruses. However, wild-type (normal) reverse transcriptase and integrase would still allow the DNA segment to move throughout the host genome, converting the active retroviral DNA into an *autonomous retrotransposon* (**Choices A and B**).

**(Choice D)** Because retrotransposons move via RNA intermediates, the sequences *must* be initially transcribed by RNA polymerase. Accordingly, DNA segments that are not transcribed into mRNA are *not* classified as retrotransposons.

**Educational objective:**

Fragments of DNA that can move between different areas of the genome are called transposons. Retrotransposons move via an RNA intermediate that is transcribed from DNA, translated into specialized enzymes (ie, reverse transcriptase, integrase), converted back to DNA, and reinserted into a new genomic location.

**Time spent:**0 sec**Subject:**

Biology

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Cellular Biology